Antitumour effects of Phyllanthus emblica L.: induction of cancer cell apoptosis and inhibition of in vivo tumour promotion and in vitro invasion o...

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Phyllanthus emblica Linn. (PE) is a medicinal fruit used in many Asian traditional medicine systems for the treatment of various diseases including cancer. The present study tested the potential anticancer effects of aqueous extract of PE in four ways: (1) against cancer cell lines, (2) in vitro apoptosis, (3) mouse skin tumourigenesis and (4) in vitro invasiveness. The PE extract at 50-100 microg/mL significantly inhibited cell growth of six human cancer cell lines, A549 (lung), HepG2 (liver), HeLa (cervical), MDA-MB-231 (breast), SK-OV3 (ovarian) and SW620 (colorectal). However, the extract was not toxic against MRC5 (normal lung fibroblast). Apoptosis in HeLa cells was also observed as PE extract caused DNA fragmentation and increased activity of caspase-3/7 and caspase-8, but not caspase-9, and up-regulation of the Fas protein indicating a death receptor-mediated mechanism of apoptosis. Treatment of PE extract on mouse skin resulted in over 50% reduction of tumour numbers and volumes in animals treated with DMBA/TPA. Lastly, 25 and 50 microg/mL of PE extract inhibited invasiveness of MDA-MB-231 cells in the in vitro Matrigel invasion assay. These results suggest P. emblica exhibits anticancer activity against selected cancer cells, and warrants further study as a possible chemopreventive and antiinvasive agent.